

Fourier Series and the Dirichlet Problem

Howell & Mathews Complex Analysis

Overview

- ▶ Fourier series representation of periodic functions
- ▶ Euler formulas for coefficients
- ▶ Harmonic functions in the unit disk
- ▶ Solving the Dirichlet problem via Fourier series
- ▶ Poisson integral formula

Periodic Functions

Let $f(\theta)$ be 2π -periodic.

Goal: represent f as a trigonometric series:

$$f(\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(n\theta) + b_n \sin(n\theta))$$

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Key idea: orthogonality of $\sin(n\theta)$ and $\cos(n\theta)$.

Orthogonality Relations

For integers $m, n \geq 0$:

$$\int_{-\pi}^{\pi} \cos(m\theta) \cos(n\theta) d\theta = \begin{cases} 0 & m \neq n \\ \pi & m = n \neq 0 \end{cases}$$

$$\int_{-\pi}^{\pi} \sin(m\theta) \sin(n\theta) d\theta = \begin{cases} 0 & m \neq n \\ \pi & m = n \end{cases}$$

$$\int_{-\pi}^{\pi} \sin(m\theta) \cos(n\theta) d\theta = 0$$

Euler Formulas

Using orthogonality:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(\theta) \cos(n\theta) d\theta$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(\theta) \sin(n\theta) d\theta$$

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$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(\theta) d\theta$$

These are called the **Euler formulas**.

Complex Form of Fourier Series

Using $e^{in\theta} = \cos(n\theta) + i \sin(n\theta)$:

$$f(\theta) = \sum_{n=-\infty}^{\infty} c_n e^{in\theta}$$

where

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta) e^{-in\theta} d\theta$$

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This form is especially natural in complex analysis.

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$$\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{in\theta} \overline{e^{im\theta}} d\theta = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(n-m)\theta} d\theta = \begin{cases} 1 & n = m \\ 0 & n \neq m \end{cases}$$

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Thus $\{e^{in\theta}\}$ forms an orthonormal basis in $L^2[-\pi, \pi]$.

This gives the coefficient formula:

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta) e^{-in\theta} d\theta$$

The Dirichlet Problem

Given a continuous function $f(\theta)$ on the boundary:

$$|z| = 1$$

Find a function $u(r, \theta)$ such that:

- ▶ u is harmonic in $|z| < 1$
- ▶ $u(1, \theta) = f(\theta)$

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Goal: construct u explicitly.

Separation of Variables Idea

Seek solutions of Laplace's equation:

$$\Delta u = 0$$

in polar coordinates:

$$u(r, \theta) = R(r)\Theta(\theta)$$

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This leads to solutions:

$$u_n(r, \theta) = r^n \cos(n\theta), \quad r^n \sin(n\theta)$$

General Harmonic Solution

General harmonic function in disk:

$$u(r, \theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} r^n (a_n \cos(n\theta) + b_n \sin(n\theta))$$

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Compare with Fourier series of boundary data.

Matching Boundary Data

If

$$f(\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(n\theta) + b_n \sin(n\theta))$$

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This gives a harmonic extension into the disk.

Goal

We want to rewrite the harmonic extension:

$$u(r, \theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} r^n (a_n \cos(n\theta) + b_n \sin(n\theta))$$

as an integral:

$$u(r, \theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(\theta - \phi) f(\phi) d\phi$$

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So we must identify the kernel $P_r(\theta)$.

Insert Fourier Coefficients

Recall:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(\phi) \cos(n\phi) d\phi$$

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Substitute into $u(r, \theta)$:

$$u(r, \theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\phi) d\phi + \sum_{n=1}^{\infty} r^n \frac{1}{\pi} \int_{-\pi}^{\pi} f(\phi) [\cos(n\theta) \cos(n\phi) + \sin(n\theta) \sin(n\phi)] d\phi$$

Trigonometric Identity

Use identity:

$$\cos(n\theta) \cos(n\phi) + \sin(n\theta) \sin(n\phi) = \cos(n(\theta - \phi))$$

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Then:

$$u(r, \theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\phi) d\phi + \sum_{n=1}^{\infty} \frac{r^n}{\pi} \int_{-\pi}^{\pi} f(\phi) \cos(n(\theta - \phi)) d\phi$$

Factor the Integral

Factor out the integral:

$$u(r, \theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\phi) \left[1 + 2 \sum_{n=1}^{\infty} r^n \cos(n(\theta - \phi)) \right] d\phi$$

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Define:

$$P_r(\theta) = 1 + 2 \sum_{n=1}^{\infty} r^n \cos(n\theta)$$

This is the Poisson kernel (Fourier form).

Summing the Series

We now compute:

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This is a geometric series:

$$= \frac{1 - r^2}{1 - 2r \cos \theta + r^2}$$

Poisson Kernel (Closed Form)

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This is the **Poisson integral formula**.

Key Properties

- ▶ $P_r(\theta) > 0$
- ▶ $\frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(\theta) d\theta = 1$
- ▶ $P_r(\theta)$ becomes sharply peaked as $r \rightarrow 1$

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Interpretation:

$$P_r \rightarrow \delta(\theta) \quad \text{as } r \rightarrow 1$$

Poisson Kernel

The solution can be written as:

$$u(r, \theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(\theta - \phi) f(\phi) d\phi$$

where

$$P_r(\theta) = \frac{1 - r^2}{1 - 2r \cos \theta + r^2}$$

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P_r is the **Poisson kernel**.

Properties of the Poisson Kernel

- ▶ $P_r(\theta) > 0$
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Thus $u(r, \theta) \rightarrow f(\theta)$ as $r \rightarrow 1$.

Interpretation

- ▶ Fourier series gives boundary data
- ▶ Radial factor r^n ensures harmonicity
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This is a central bridge between:

- ▶ harmonic functions
- ▶ complex analytic functions
- ▶ Fourier analysis

Connection to Complex Analysis

If f is boundary values of $\Re(F)$ for analytic F :

$$F(z) = \sum_{n=0}^{\infty} c_n z^n$$

then:

$$u(r, \theta) = \Re(F(re^{i\theta}))$$

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Harmonic functions arise as real parts of analytic functions.

Summary

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Key takeaway:

Boundary data $\Rightarrow$ unique harmonic function in disk